

and hallucinations about half an hour before headache starts.

2. Ischemia sensory + low O₂ tension vasodilatation → excessive distension and stretch of cerebral arteries → severe headache.

- Alcoholic headache: Alcohol intoxication produces headache through: meningeal irritation and cerebral vasodilatation.
- Constipation: Leads to absorption of toxic products of fermented and putrefied food materials in the large intestine → meningeal irritation.

2. Physiology of Pain (I)

For scientific and clinical purposes, pain is defined by the International Association for the Study of Pain (IASP) as, “an unpleasant sensory and emotional experience associated with actual or potential tissue damage, or described in terms of such damage.” This is to be distinguished from the term nociception which the IASP defines as the unconscious activity induced by a harmful stimulus applied to sense receptors. Pain is very important warning sensation that is produced whenever any tissue is damaged.

It initiates protective reflexes for removing the injurious stimulus.

Pain receptors:

- They are free nerve endings (FNE).
- They are widely

distributed: In the skin →

cutaneous pain.

In deep tissues → deep pain.

In viscera → visceral pain.

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- They are slowly adapting, so they continuously inform the CNS about tissue damage.
- They have high threshold, so they are stimulated by very strong stimulus that produce damage to the tissue (Noxious).

The stimulus:

Any form of stimulus (mechanical, thermal, chemical,...etc) provided it is too strong to produce tissue damage (Noxious).

N.B.:

* Mechanical and thermal stimuli usually produce reflex withdrawal to protect the limb.

* Chemical stimuli produce immobilization of the limb:

1. To localize the damage.
2. To promote healing.

2. Mechanism of stimulation:

3. Tissue damage → release of products called pain producing substances (PPS) as ATP, K⁺, Bradykinin, Leukotriens, Prostaglandins, Histamine, Serotonin and substance P (Lewis's p factor) which depolarize and stimulate pain receptors.

Qualities (Characters) of Pain:

1. Pricking pain: as when the skin is pricked by a pin or a knife. It is a highly localized type of pain.

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2. Burning pain: as when the skin is burned. It is diffuse type of pain.
3. Aching pain: it is a deep and diffuse type of pain and usually referred.
4. Colicky pain: it is due to spasm of hollow viscera.

(I) CUTANEOUS PAIN (CP):

It is perceived when pain receptor in the skin are stimulated.

Clinical tests:

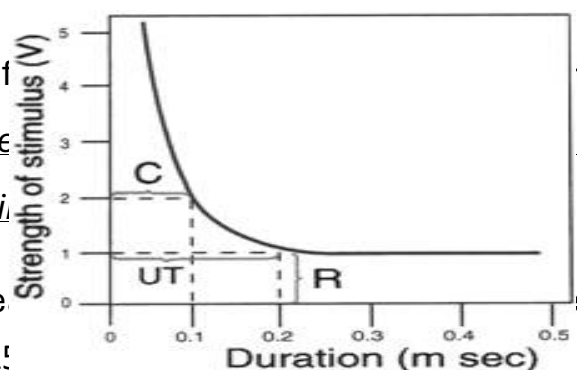
Pricking Method:

- Ask the patient to close his eyes.
- Prick the skin with a pin at equal pressure and ask the patient what he feels?.
- Corresponding points on both sides, as well as, the upper and lower limbs are compared.
- Equal sensation of pain should be perceived from corresponding points.

Heat Method:

- It is an experimental method.
- The stimulus: is heat produced by light rays emitted from an electric lamp and concentrated on a specific point on the body by means of converging lens. The intensity of heat can be controlled by an adjustable electric resistance.
- The duration of application for each strength is measured and a strength- duration curve is drawn.

- The threshold for pain : pain is felt at 45 °C in all individuals. The threshold for pain is 45 °C in all individuals. The threshold for pain is 45 °C in all individuals. The threshold for pain is 45 °C in all individuals.
- The higher is the intensity of heat, the shorter is the duration of application.



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time needed to feel pain sensation (figure 16).

Figure 16 strength duration curve

For pain sensation

Qualities (Types) of CP Sensation:

When a sudden injurious stimulus is applied to the skin, two types of pain sensations are perceived:

- A fast pricking pain sensation (immediate pain).
- A slow burning pain (delayed).

Differences between immediate and delayed pain (table 4):

	Immediate pain	Delayed pain
Time	-Felt during injury and of short duration	-Felt shortly after injury and of long duration
Character	-Sharp localized pricking	-Diffuse burning
Afferent fibers	-Thin A δ fibers. The rate of conduction is 5-30 m/s	-Nonmyelinated C fibers. The rate of conduction is 0.5-2 m/s
Clinical significance	-Rapidly inform the CNS about the injurious stimulus. -Initiates protective withdrawal reflexes.	-Can summate and becomes intolerable.
Dissociation of the two types: Effect of afferent nerve trunk compression	-Destroys A δ fibers before C fibers, so immediate pain disappears first.	Burning pain persists longer and disappears at the end. Burning pain disappears first.

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Effect of local anesthetics	Blocks the C fibers before A δ fibers, so immediate pain persists longer and disappears at the end	
Center	The cerebral cortex	The thalamus

Pathway of pain sensation:

The first order neuron:

Cutaneous pain is conducted from pain receptors in the skin (FNE) along two types of afferent fibers:

1. Thin myelinated A δ fibers responsible for immediate pain sensation.
2. C nonmyelinated fibers responsible for delayed burning pain sensation.

The cell bodies of either types are present in DRG and they enter the spinal cord along the lateral division of the dorsal root to ascend or descend for one or two segments forming short Lissauer's tract to end on the neurons of the SGR (Lamina II & III) at the tip of the dorsal horn gray matter of the spinal cord through several ICN with multiple synapses acting as gates. There is evidence that the fibers serving mild pain secrete glutamate transmitter, while those serving severe pain secrete substance p.

The second order neuron:

Axons of SGR cross the middle line in front of the central canal of the spinal cord and ascend in the lateral white matter of the spinal cord of the opposite side as the lateral spinothalamic tract (LST) to the brain stem.

In the brain stem, the two pathways for immediate and delayed pain separate as follows:

Pathway for immediate pain (figure 17):

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The neurons of the LST conducting immediate pain end on PLVNT.

The third order neuron:

- Axons of PLVNT ascend in the posterior limb of IC to form the sensory radiation to end in the sensory area of the CC in the postcentral gyrus.

Pathway for delayed pain:

- The neurons of the LST conducting delayed pain end on the intralaminar thalamic nuclei and nuclei of the reticular formation.

The third order neuron:

- Axons of these nuclei discharge to all areas of the CC leading to alertness and anxiety.
- The thalamus is a higher center that can perceive pain sensation irrespective of the CC.
- The CC helps localizing the site of pain sharply.

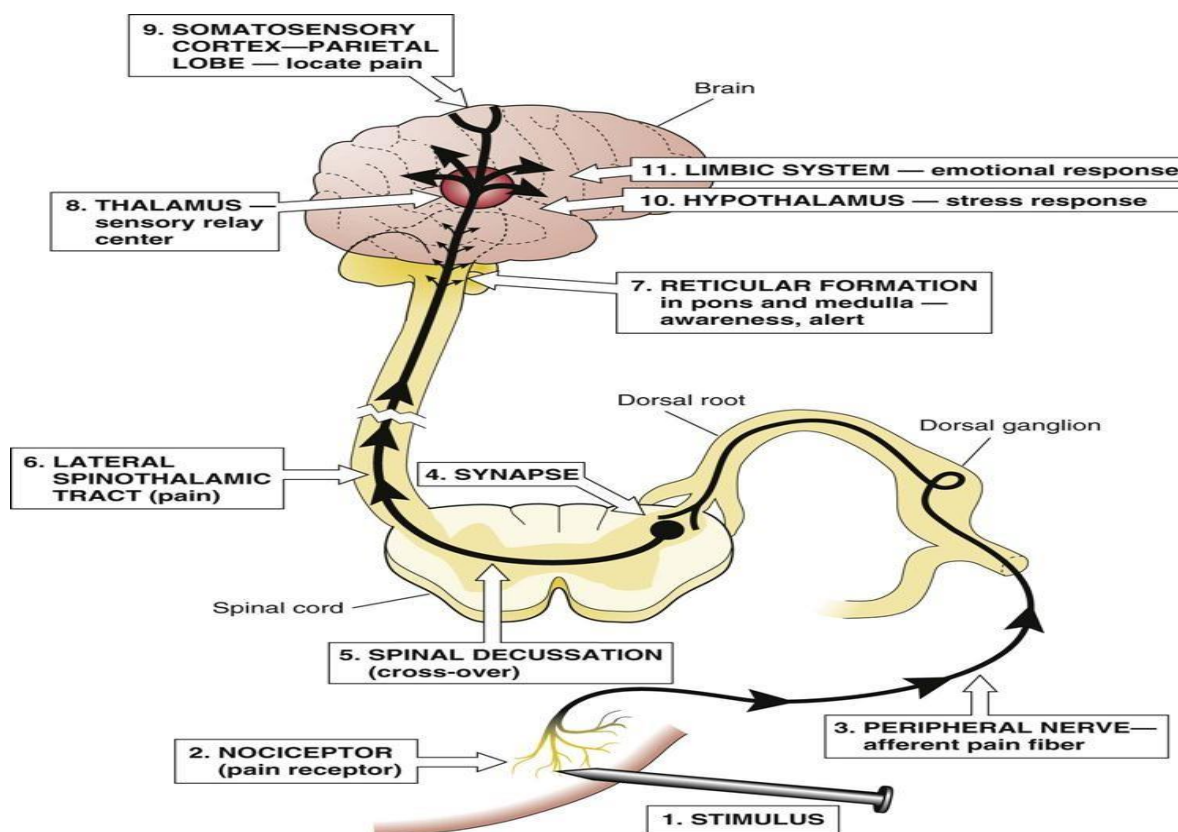


Figure 17: immediate pain pathway

Reactions of the body to pain:

(1) Somatic Reflexes

(table They are (table 5):

1 Immobilization reflex	2 Flexion withdrawal reflex
The stimulus: usually chemical injurious stimuli	The stimulus : usually mechanical or thermal injurious stimuli
Function: localize the damage and promote healing	Function: withdraw the limb from injurious stimuli

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Characters:

There centers are in spinal cord.

They are protective

The flexion withdrawal is prepotent i.e. can inhibit all other reflexes occurring at the same time.

Pathway:

The afferent impulses carrying pain enter the cord through the posterior root and pass through collaterals that end on the AHC supplying the effector organ

(2) Autonomic Reflexes

Depending on the severity of pain:

- Moderate pain → sympathetic response in the form of tachycardia, increased blood pressure,...etc.
- Severe pain → parasympathetic response in the form of bradycardia, decreased blood pressure, vomiting,...etc.
- Mechanism : collateral branches from LST conducting pain impulses relay on and stimulate autonomic centers in the medulla and reticular formation like the cardiac, vomiting, ...etc → autonomic reflexes.

(3) Emotional Reactions

All individuals have the same threshold for pain, but they differ in their emotional reaction (behavior) to pain which may take the form of crying, weeping, anxiety,...etc.

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The pain impulses serving emotional reactions should be relayed in CINGULATE GYRUS.

(4) Hyperalgesia:

- Definition: a pathological skin condition, in which pain can be produced by non noxious stimuli which under normal conditions are not painful.
- Causes:
 1. Inflammatory skin condition.
 2. Irritation of afferent fibers or DRG.
 3. Occasionally associated with deep visceral diseases.
 4. Time of occurrence: It occurs few minutes after injury (20-30 minutes) and persists for several hours or days later.

Types of Hyperalgesia (table 6):

	Primary hyperalgesia	Secondary hyperalgesia
Site	All the site of skin area showing redness (Red Flare)	In healthy skin around the injured area.
Pain threshold	Lowered	Not lowered and may be increased
Central response	Intense feeling of pain	Intense feeling of pain

Mechanism (table 7)

Primary hyperalgesia	Secondary hyperalgesia
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1-Skin damage → release of PPS from the damaged cells → diffuse in the damaged area → lower the threshold of pain receptors → hyperalgesia. 2-Local axon reflexes: skin damage → afferent discharge of pain impulses → pass to CNS → pain sensation. Some impulses pass through branches of afferent fibers back to the area of skin damage (antidromic conduction) to end on : i-Skin arterioles → VD (Red Flare). ii-Cutaneous cells (mast cells) → release of PPS → lower the threshold of pain receptors in the area → hyperalgesia.	1-Cortical neurons for the area of secondary hyperalgesia are laterally inhibited by collateral neurons from area of primary hyperalgesia for the sake of sharp localization, hence the threshold of pain in area of secondary hyperalgesia is increased. 2-Convergence-facilitation theory: it explains why the response for pain from the area of secondary hyperalgesia is more intense than normal. -Afferents from area of primary hyperalgesia (through collaterals) converge on the same central neuron (SGR, Thalamus or CC) for the area of secondary hyperalgesia increasing their excitability
iii-The hyperexcitable pain receptors discharge at a higher rate and the feeling of pain becomes intense.	(facilitation) → intense pain response.

3. Physiology of Pain (II)

Control of pain sensation:

Pain pathway can be inhibited as evidenced by:

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- Stress analgesia: Soldiers wounded in the battle field may feel no pain until the battle is over.
- Rubbing of the skin around the injured area decrease the degree of pain sensation.
- Morphine is used as a powerful analgesic drug to reduce pain sensation.

GATE THEORY:

Mechanism:

Control of pain depends on the GATE THEORY FOR PAIN as follows: The relay between the first and second order neurons conducting pain impulses at the SGR involves a chain of several short ICN in between. The impulses should pass through several synapses or gates. These gates offer resistance to conduction of pain impulses and are under control from spinal and supraspinal pathways as follows:

Spinal And Supraspinal Pathways:

1) Spinal control of pain sensation:

The afferent nonmyelinated C fibers carrying impulses for delayed burning pain sensation activate the neurons of DRG (2nd order neurone through the release of a chemical transmitter (substance p).

Collaterals from large myelinated A β fibers that enter the dorsal column carrying tactile sensory impulses pass to end on the termination of the C fibers for pain and produce presynaptic inhibition $\rightarrow \downarrow$ the release of substance p $\rightarrow \downarrow$ pain transmission at the SGR $\rightarrow \downarrow$ pain sensation.

Clinical Significance:

- This explains why rubbing the skin around the injured area decreases